Unsupervised Learning

Clustering: a technique where data can be divided into a number of Logical groups based on the characteristics of the data

⇒ goal of clustering algorithm is to ensure that all of the data points within the same cluster should be as same as possible.

⇒ K-means clustering Algorithm :-

↳ unsupervised ML Algo. , Groups unlabeled data points into K distinct clusters based on Similarity

↳ Centroid → the center point of a cluster

⇒ Steps of K-means Algorithm :-

1. Choose K : decide the number of clusters (ex K=2)
2. Random initialization : place K centroids randomly in the data space

3. Assign each data point to its nearest centroid using distance Formulas.

Euclidean Distance $= \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$

distance between each point and centroid

4. Update Centroids: Calculate the mean "avg" of all data points in each cluster and move the centroid to this new average position

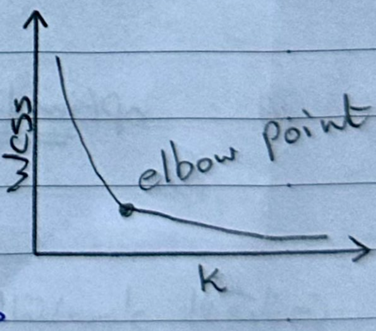
5. Repeat until centroids stop changing (convergence).

* How to choose the Best K??

↳ (The Elbow Method):-

↳ WCSS → within-cluster sum of squares

↳ sum of squared distances between data points and thier assigned cluster centroid.



↳ Steps:

1. Run K-means for $K = 1 \rightarrow 20$
2. Compute WCSS for each K
3. plot line chart of K Vs WCSS
4. Look for the "Elbow Point" → the point where WCSS decreases rapidly and then levels off

→ the K value with elbow point is the optimal K.

التاريخ: / / الموضوع:

2> Inliers: A point perfectly centered in its own cluster

2> outliers: A point on the edge or mistakenly placed near a different cluster.

التقارب الداخلي

2> a "Cohesion": Average distance to all other points in the same cluster
(Smaller = Better)

التباعد الخارجي

2> b "separation": Average distance to points in the Next closest cluster
(Larger = Better)

Silhouette Score.

خوارزمية بتقيس مدى تماسك العناصر جوا نفس الـ cluster مقارنة بمدى انعزالها
عن باقي الـ clusters التانية في نفس الـ dataset

← بنستخدمه مع الـ K-means حيث أعلى قيمة لـ S تدي optimal K
يعني بنعتبره طريقة Validation

← بنعتبر كانه مؤشر للتباين كل ما كانت الـ clusers متباعدة او الاختلاف بينها اكبر دة أفضل وبنوصل للنتيجة دي عن طريق انه
(a_i ↓ معدل التقارب
و b_i ↑ معدل التباعد)

⇒ formula :-

Silhouette Score : $s = \dfrac{b-a}{\max(a,b)}$

↳ s Value is between $[-1, 1]$

⇒ Near (+1) : perfect inlier "Right cluster".

النقاط متقاربة ومتشابهة لبعضها و مختلفة عن المجموعات اللي حواليها

⇒ Near (0) : Borderline point (on the boundry)

بعض النقط تقع على الحدود الفاصلة بين مجموعتين و في احتمال تداخل

⇒ Near (-1) : outlier (wrong cluster , closer to the neighbors)

في نقط اتحطت في مجموعة غلط لأنها أقرب لمجموعة تانية

⇒ Use Case : Finding outliers in 2D is easy , but in High Dimensions " complex Data" , our eyes fail and so the Silhouette score mathematically detect them

Principal Component Analysis (PCA)

→ PCA: is a data compression tool, it takes complex data and compresses it into smaller number of columns called "Principal Components or PCs" while keeping the most important information and deleting usless noise

→ How PCA works:

⇒ line and Angle (direction):

data points scattered on a 2D graph, PCA draws a straight line through the middle of the points and projects every single point onto this line at a (90°) angle

⇒ Rotating the line:

PCA rotates this line in all possible directions searching for one specific Maximum Variation (spread)

⇒ PC1 (First principal component)

→ the line (direction) where the ~~w~~ projected points are spread out the widest, meaning it holds the highest amount of information and data spread

الموضوع: التاريخ: / /

PC2: the next best direction that holds the second-highest amount of information.
↳ with one strict rule: it must be orthogonal to PC1
PC_2 عمودي على PC1

⇒ PCA Uses Correlation Metrix to Combine Similar Variables

⇒ Eigen Vectors: gives us the direction of PCs

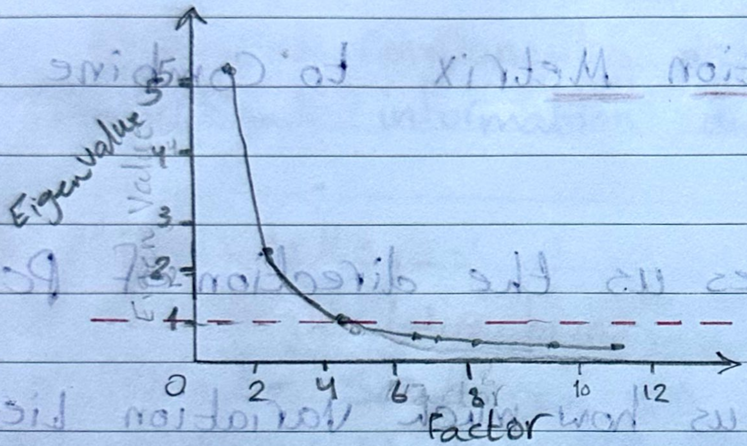
⇒ Eigen Values: Shows us how much variation lies along each ~~of~~ those directions

Important Note: "The Scale problem (standardization)"
↳ if we have one variable with huge number and another with small numbers, PCA will mistakenly think that the large numbers is more important
↳ in order to fix that: we must always apply Standardization (scalling) before running PCA so that every Feature starts on an equal scale

⇒ Explained Variance table: shows us the percentage of total information captured by each PC

⇒ Scree plot : An elbow-shaped Graph used to decide when to stop taking extra PC

↳ we keep what is before the elbow drop and ignore the rest



⇒ Kaiser Criterion : altenative of Scree plot keeps only PCs that ~~keeps~~ have an EigenValue greater than 1.

التاريخ: / / الموضوع:

⇒ Evaluating Clustering Models :

إنه أقول إن الـ clusters اللي معايا perfect

if we have 2 features, we can look at the graph and see if the clusters are good, but with High Dimensions we can't visualize them

↳ and so we use mathematical metrices

⇒ **Intra-cluster** : distance (inside) between points inside the same Cluster and centroide

↳ goal : minimum intra-cluster distance so the

⇒ Points are tightly packed together

⇒ **Inter-cluster distance** : distance between two different clusters

↳ goal : Maximum "clusters' far away from each other

⇒ Metric 1 : Inertia

↳ calculates the sum of all intra-cluster distances, it measures how close the points in a cluster are to thier center

↳ less Inertia the Better cluster

inertia مش أفضل evaluating Metix عشان مش بتعمل حساب اللي برا الـ cluster

وكل الشغل فيها جوا الـ cluster

⇒ Metric 2 : Dunn ~~cluster~~ index

↳ Dunn index = $\dfrac{\text{min(Inter cluster distances)}}{\text{max(Intra cluster distances)}}$

the Higher Dunn Index = Better clusters